

Derivation of the Green's Function for the Heat Equation

The Heat Equation

We consider the initial value problem

$$u_t = ku_{xx}, \quad x \in \mathbb{R}, \quad t > 0$$

with initial condition

$$u(x, 0) = f(x).$$

Goal: express the solution in terms of the initial data.

We will construct a **Green's function** for the heat operator.

The Green's Function Problem

We seek a function

$$G(x, t; \xi)$$

satisfying

$$G_t = kG_{xx}, \quad t > 0$$

with initial condition

$$G(x, 0; \xi) = \delta(x - \xi).$$

Interpretation:

- ▶ initial heat concentrated at the point ξ
- ▶ $G(x, t; \xi)$ describes how that heat spreads

Translation Invariance

The heat equation on $\mathbb{R}$ is translation invariant.

Therefore the Green's function depends only on the difference

$$x - \xi.$$

We write

$$G(x, t; \xi) = K(x - \xi, t).$$

Thus K satisfies

$$K_t = kK_{xx}$$

with

$$K(x, 0) = \delta(x).$$

The Fourier Transform

To solve the PDE we apply the Fourier transform in the spatial variable.

For a function $u(x)$ we define

$$\hat{u}(p) = \int_{-\infty}^{\infty} u(x) e^{-ipx} dx.$$

The inverse transform is

$$u(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{u}(p) e^{ipx} dp.$$

The Fourier transform decomposes a function into its **frequency components**.

Fourier Transform of Derivatives

A key property is how derivatives transform.

If

$$\hat{u}(p) = \mathcal{F}[u](p),$$

then

$$\mathcal{F}[u_x](p) = ip \hat{u}(p),$$

and

$$\mathcal{F}[u_{xx}](p) = -(p^2) \hat{u}(p).$$

Thus differentiation in x becomes multiplication in the frequency variable p .

Symbols of Differential Operators

Let L be a constant-coefficient differential operator

$$L = a_0 + a_1 \partial_x + a_2 \partial_{xx} + \cdots .$$

Applying the Fourier transform gives

$$\mathcal{F}[Lu](p) = \sigma_L(p) \hat{u}(p).$$

The polynomial

$$\sigma_L(p)$$

is called the **symbol** of the operator.

It is obtained by replacing

$$\partial_x \rightarrow ip.$$

Symbol of the Heat Operator

Consider the heat operator

$$L = \partial_t - k\partial_{xx}.$$

Taking the Fourier transform in x gives

$$\partial_t \hat{u}(p, t) = -kp^2 \hat{u}(p, t).$$

Thus the spatial part of the operator has symbol

$$\sigma(p) = -kp^2.$$

The PDE therefore reduces to an ODE for each frequency.

Why the Method Works

The Fourier transform diagonalizes constant-coefficient differential operators.

$$u(x, t) = \int \hat{u}(p, t) e^{ipx} dp$$

Each Fourier mode evolves independently according to

$$\frac{d}{dt} \hat{u}(p, t) = \sigma(p) \hat{u}(p, t).$$

For the heat equation

$$\sigma(p) = -kp^2$$

so high frequencies decay rapidly.

Fourier Transform

Take the Fourier transform in x :

$$\hat{K}(p, t) = \int_{-\infty}^{\infty} K(x, t) e^{-ipx} dx.$$

Using

$$\mathcal{F}(K_{xx}) = -(p^2) \hat{K},$$

the PDE becomes

$$\frac{\partial}{\partial t} \hat{K}(p, t) = -kp^2 \hat{K}(p, t).$$

Solving the Transformed Equation

For each p we obtain the ODE

$$\frac{d}{dt}\hat{K}(p, t) = -kp^2\hat{K}(p, t).$$

Solution:

$$\hat{K}(p, t) = C(p)e^{-kp^2t}.$$

The initial condition determines $C(p)$.

Initial Condition

Since

$$K(x, 0) = \delta(x),$$

its Fourier transform is

$$\hat{K}(p, 0) = 1.$$

Thus

$$C(p) = 1.$$

Therefore

$$\hat{K}(p, t) = e^{-kp^2t}.$$

Inverse Fourier Transform

Recover $K(x, t)$ using the inverse transform:

$$K(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-kp^2t} e^{ipx} dp.$$

This is a Gaussian integral.

Evaluating it gives

$$K(x, t) = \frac{1}{\sqrt{4\pi kt}} \exp\left(-\frac{x^2}{4kt}\right).$$

The Heat Kernel

Thus the Green's function is

$$G(x, t; \xi) = \frac{1}{\sqrt{4\pi kt}} \exp\left(-\frac{(x - \xi)^2}{4kt}\right).$$

This function is called the **heat kernel**.

Properties:

- ▶ $G_t = kG_{xx}$
- ▶ $G(x, 0; \xi) = \delta(x - \xi)$
- ▶ total heat is conserved

Constructing the Solution

Using superposition,

$$f(x) = \int_{-\infty}^{\infty} f(\xi) \delta(x - \xi) d\xi.$$

Replacing the delta by the Green's function gives

$$u(x, t) = \int_{-\infty}^{\infty} G(x, t; \xi) f(\xi) d\xi.$$

Final Solution Formula

Substituting the heat kernel,

$$u(x, t) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi kt}} \exp\left(-\frac{(x - \xi)^2}{4kt}\right) f(\xi) d\xi.$$

Thus the solution is the convolution

$$u(x, t) = (K(\cdot, t) * f)(x).$$

Interpretation

The heat kernel describes diffusion of an initial point source.

- ▶ Initially all heat is at one point.
- ▶ As time increases the distribution spreads.
- ▶ The Gaussian width grows like $\sqrt{t}$.

Every solution is a superposition of these spreading point sources.